NEWTON SCHOOL

PRE-CALCULUS

Inverse Trig Evaluation: Real Numbers (Degrees & Radians)

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Standard: *CCSS.MATH.CONTENT.HSF-BF.B.4 — Find inverse functions. Evaluate* $\sin^{-1}$, $\cos^{-1}$, *and* $\tan^{-1}$ *of standard unit-circle values and express answers in both degree and radian measure.*

Quick Reference — Unit Circle (First Quadrant + Axis)

Angle	$\sin\theta$	$\cos\theta$	$\tan\theta$
$0^\circ = 0$	0	1	0
$30^\circ = \frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
$45^\circ = \frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
$60^\circ = \frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$90^\circ = \frac{\pi}{2}$	1	0	undef.

🔔 Mr. Kenny's Key Point

For every answer, give **both degrees and radians** — fluency in both is essential for Pre-Calc.

Stay inside each function's range: $\sin^{-1} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ · $\cos^{-1} \in [0, \pi]$ · $\tan^{-1} \in (-\frac{\pi}{2}, \frac{\pi}{2})$

Part A — Evaluate $\sin^{-1}$ (7 problems)

Directions: *Evaluate each inverse sine expression. Write your answer in* **both** *degrees and radians in the spaces provided.*

1. $\sin^{-1}\left(\frac{1}{2}\right) =$

Degrees: ________________ *Radians:* ________________

2. $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) =$

Degrees: ________________ *Radians:* ________________

3. $\sin^{-1}\left(-\frac{1}{2}\right) =$

Degrees: ________________ *Radians:* ________________

4. $\sin^{-1}(0) =$

Degrees: ______________ *Radians:* ______________

5. $\sin^{-1}\left(\frac{\sqrt{2}}{2}\right) =$

Degrees: ________________ *Radians:* ________________

6. $\sin^{-1}(-1) =$

Degrees: ________________ *Radians:* ________________

7. $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) =$

Degrees: ________________ *Radians:* ________________

Part B — Evaluate $\cos^{-1}$ (7 problems)

Directions: *Evaluate each inverse cosine expression. Write your answer in* **both** *degrees and radians. Note:* $\cos^{-1}$ *range is* $[0^\circ, 180^\circ]$.

8. $\cos^{-1}(0) =$

Degrees: ________________ *Radians:* ________________

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9. $\cos^{-1}\left(\frac{1}{2}\right) =$

Degrees: ________________ *Radians:* ________________

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10. $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) =$

Degrees: ________________ *Radians:* ________________

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11. $\cos^{-1}\left(-\frac{1}{2}\right) =$

Degrees: ________________ *Radians:* ________________

12. $\cos^{-1}\left(\frac{\sqrt{2}}{2}\right) =$

Degrees: ________________ *Radians:* ________________

13. $\cos^{-1}(-1) =$

Degrees: ________________ *Radians:* ________________

14. $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) =$

Degrees: ________________ *Radians:* ________________

Part C — Evaluate $\tan^{-1}$ (6 problems)

Directions: *Evaluate each inverse tangent expression. Write your answer in* **both** *degrees and radians. Note:* $\tan^{-1}$ *range is* $(-90^\circ, 90^\circ)$.

15. $\tan^{-1}(1) =$

Degrees: ______________ *Radians:* ______________

16. $\tan^{-1}(\sqrt{3}) =$

Degrees: ______________ *Radians:* ______________

17. $\tan^{-1}(0) =$

Degrees: ______________ *Radians:* ______________

18. $\tan^{-1}(-1) =$

Degrees: ______________ *Radians:* ______________

19. $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) =$

Degrees: ________________ *Radians:* ________________

20. $\tan^{-1}(-\sqrt{3}) =$

Degrees: ________________ *Radians:* ________________